



EXECUTION PLAN

VIRTUAL CONTROL
VCF NETWORK RECOVERY ENGINEERING

Management Plane vDS Migration

Move four VCF management VMs from the orphan vSphere Distributed Switch onto the live vcenter-cl01-vds01 ephemeral and static port groups, closing VCFA pre-deploy gates HB-1 and HB-2 and removing the last live references that have blocked NSX cleanup.

VDS MIGRATION

EPHEMERAL BINDING

VCENTER MOVE

HB-1 HB-2

KB 426300

VCF 9.0.1

vSphere 9.0 • NSX 9.0

VCF Management Plane vDS Migration Plan

Prepared by: Virtual Control LLC **Date:** May 16, 2026 **Document Version:** 1.0 **Classification:** Internal — VCF Operations **Scope:** Move four management VMs (`sddc-manager` , `vcf-ops` , `fleet` , `vcenter`) off the orphan vDS (UUID `50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8`) and onto live vDS `vcenter-cl01-vds01` (UUID `50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf`). Closes VCFA-pre-deploy gates HB-1 and HB-2, and removes the last live VM references to the orphan vDS so that NSX `nsx-ownership clear` , TNC detach, and per-host TN reset paths (which all failed earlier today against three different NSX guard rails) can finally complete.

Table of Contents

- 1. Executive Summary
- 2. Context
- 3. Verified Facts
- 4. Open Decisions
- 5. Phase 0 — Pre-flight Verification
- 6. Phase 1 — Migrate `sddc-manager`
- 7. Phase 2 — Migrate `vcf-ops`
- 8. Phase 3 — Migrate `fleet`
- 9. Phase 4 — Migrate vCenter (Special Case)
- 10. Phase 5 — Post-Migration Verification
- 11. Phase 6 — Orphan vDS Removal (Optional)
- 12. Risk Register
- 13. Rollback Procedures
- 14. KB References
- Appendix A — Commands

1. Executive Summary

Diagnostic work on 2026-05-16 surfaced that the cluster `vcenter-cl01` runs **two parallel vSphere Distributed Switches** — the live `vcenter-cl01-vds01` (UUID `50 12 35 19...`) and an orphan vDS (UUID `50 37 13 3f...`) that had been presumed deleted. Four critical management VMs remain attached to the orphan vDS:

VM	Current DVPG	vDS	DVPG Inventory Status
<code>sddc-manager</code>	<code>dvportgroup-59</code>	orphan	Stale — DVPG name not resolvable
<code>vcf-ops</code>	<code>dvportgroup-59</code>	orphan	Stale
<code>fleet</code>	<code>dvportgroup-59</code>	orphan	Stale
<code>vcenter</code>	<code>dvportgroup-6025</code>	orphan	Deleted from inventory — port persists
<code>nsx-manager</code>	<code>dvportgroup-49</code> (<code>pg-mgmt</code>)	live	OK
<code>collector</code>	<code>dvportgroup-49</code> (<code>pg-mgmt</code>)	live	OK

This is the proximate cause of three downstream NSX cleanup failures observed earlier today (error_code 8012 clearing NSX ownership, 9560 on per-host TN PUTs, 26185 on TNC DELETE). With four live VM references still bound to the orphan vDS, NSX cannot release ownership and vCenter cannot delete the orphan switch.

This plan migrates the four VMs to live-vDS port groups in least-risk order. vCenter, being the manager performing its own NIC change, follows the documented two-NIC procedure per [BROADCOM-DOC: KB 418461] to avoid the well-known self-disconnect failure mode.

Outcome on success: HB-1 and HB-2 satisfied. Orphan vDS has zero VM references and becomes deletable. NSX ownership-clear, per-host TN reset, and TNC detach paths all unblock. VCFA pre-deploy gates 1-2 complete.

2. Context

2.1 How the discovery was made

Get-NetworkAdapter on vcenter returned:

```
PortgroupKey : dvportgroup-6025
SwitchUuid   : 50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8
```

Cross-checking via Get-View -Id "DistributedVirtualPortgroup-dvportgroup-6025" returned View ... was not found on the server(s) — the DVPG no longer exists in inventory, yet vCenter's vNIC retains a live port connection to it. The live vDS shows DVPG keys dvportgroup-44 through dvportgroup-50 only.

A full sweep of all VMs revealed three more on the same orphan switch UUID (sddc-manager , vcf-ops , fleet — all on dvportgroup-59).

2.2 Why this blocks VCFA deploy

The VCFA Deploy Airtight Plan dated 2026-05-12 lists two hard blockers driven by Broadcom requirements:

- **HB-1** — vCenter must be on a DVS port group [BROADCOM-DOC: KB 420920, KB 430625] . Current state: vCenter is technically on a DVS port group, but the DVPG no longer exists in inventory. This is structurally fragile — any vSphere garbage-collection of the orphan switch would sever vCenter's connectivity.
- **HB-2** — Management port group must use ephemeral binding [BROADCOM-DOC: KB 324492, KB 426300, KB 428279] . Current state: cannot be verified (DVPG missing); even if the orphan DVPG was static, vCenter is at risk during any future migration because static binding requires vCenter itself to assign the port — the very vCenter being moved [BROADCOM-DOC: KB 418461] .

2.3 Why this blocks NSX cleanup

The NSX nsx-ownership table records that the orphan vDS UUID is "owned by NSX" on Compute Manager 259b8edb NSX's documented clear action calls back to vCenter to release the ownership; vCenter responds "no such vDS" (because the switch is technically still there but the port group reference is dead). NSX has no force=true option (verified — action enum is locked to [set, clear] , error_code 255). TNC DELETE is gated by lifecycleManaged=true on the compute collection (error_code 26185). Per-host TN PUT is gated by transport-zone single-transaction relocation rule (error_code 9560).

All three gates have the same underlying cause: the orphan vDS has live VM references, so NSX (correctly) refuses to release it.

Once these four VMs migrate off the orphan vDS, the orphan can be removed from vCenter cleanly, and the NSX cleanup paths will work as documented.

3. Verified Facts

All values below are confirmed on 2026-05-16 via PowerCLI / NSX API; nothing is inferred.

3.1 Live vDS (target)

Property	Value	Source
Display name	vcenter-cl01-vds01	Get-VDSwitch
UUID	50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf	Get-VDSwitch.ExtensionData.Uuid
Version	9.0.0	Get-VDSwitch.Version
Hosts attached	4 (esxi01-04)	NSX TN realization

3.2 Live vDS port groups

Display Name	MoRef Key	PortBinding	Role
vcenter-cl01-vds-DVUplinks-43	dvportgroup-44	Static	Uplinks (do not use for VMs)
pg-mgmt	dvportgroup-45	Static	Target for sddc-manager, vcf-ops, fleet
pg-vmotion	dvportgroup-46	Static	vMotion
pg-vsan	dvportgroup-47	Static	vSAN
pg-overlay-tep	dvportgroup-48	Static	NSX overlay TEP
pg-vm-mgmt	dvportgroup-49	Static	Workload management (already used by nsx-manager, collector)
pg-mgmt-ephemeral	dvportgroup-50	Ephemeral	Target for vcenter

3.3 Orphan vDS (source — to be evacuated)

Property	Value
UUID	50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8
Stale DVPG keys still referenced	dvportgroup-59 (3 VMs), dvportgroup-6025 (1 VM, deleted from inventory)
NSX TZ realization still attached	TZ 3fbf2d31-12a9-4781-8cd3-95b4bf472824 on all 4 TNs

Property	Value
NSX ownership record	Compute Manager 259b8edb-3b54-496f-bfb4-c85c145cd9bb → orphan vDS

3.4 VMs to migrate

VM	Hosted On	Current NIC State	Target
sddc-manager	(verify per-VM)	dvportgroup-59 (orphan)	pg-mgmt
vcf-ops	(verify per-VM)	dvportgroup-59 (orphan)	pg-mgmt
fleet	(verify per-VM)	dvportgroup-59 (orphan)	pg-mgmt
vcenter	esxi04.lab.local	dvportgroup-6025 (orphan, stale)	pg-mgmt-ephemeral

3.5 Cluster state at plan creation

Item	Value
NSX manager cluster	STABLE
vSAN resync remaining	0 GB, 0 components
DRS automation	FullyAutomated
HA	Disabled (acceptable for this work)
Segments deployed	0
Hosts in MM	0

4. Open Decisions

#	Decision	Default	Notes
D 1	Migrate vCenter using dual-NIC method [BROADCOM-DOC: KB 418461] or in-place NIC swap via PowerCLI direct-to-ESXi?	Dual-NIC (KB-documented safe pattern)	Dual-NIC: add second NIC on pg-mgmt-ephemeral, verify, remove old. Safer. In-place: faster but vCenter session drops mid-operation.
D 2	Do we shut VMs down before NIC swap, or hot-swap?	Hot-swap	All 4 VMs support NIC hot-swap; the underlying VLAN is identical, so no IP renumber is needed. Validated in §3.2 — all live-vDS mgmt DVPGs use the same management VLAN as the orphan DVPGs.
D 3	After migration completes, do we delete the orphan vDS now (Phase 6) or in a separate change window?	Separate window	Phase 6 is included for completeness but is gated by user authorization separately.

5. Phase 0 — Pre-flight Verification

Goal: prove the plan's preconditions still hold at execution time. Stop and re-plan if any check fails.

5.1 Confirm orphan and live vDS UUIDs unchanged [VERIFIED 2026-05-16]

```
Connect-VIServer vcenter.lab.local
Get-VDSwitch | Select-Object Name, @{N='UUID';E={$_.ExtensionData.Uuid}}, Version
```

Gate: Output lists exactly one switch named `vcenter-cl01-vds01` with the documented UUID. If a second switch with the orphan UUID appears in `Get-VDSwitch`, the orphan vDS is still **registered** in vCenter (vs being a ghost reference) and Phase 6 will work via `Remove-VDSwitch`. If it does not appear, the orphan exists only as port-key references and Phase 6 requires manual MOB cleanup (out of scope here).

5.2 Re-enumerate VM NIC bindings [VERIFIED 2026-05-16]

```
Get-VM | ForEach-Object {
    $v = $_
    Get-NetworkAdapter -VM $v -ErrorAction SilentlyContinue | ForEach-Object {
        [PSCustomObject]@{
            VM           = $v.Name
            PortgroupKey = $_.ExtensionData.Backing.Port.PortgroupKey
            SwitchUuid   = $_.ExtensionData.Backing.Port.SwitchUuid
        }
    }
} | Format-Table -AutoSize
```

Gate: Output matches §3.4. If any of the four target VMs has already migrated (e.g., user moved one via UI) or a fifth VM appears on the orphan UUID, update §3.4 before proceeding.

5.3 Verify target DVPGs and their VLAN ID matches the orphan management VLAN

```
# Live DVPGs to which VMs will migrate
Get-VDPortgroup -Name pg-mgmt, pg-mgmt-ephemeral |
    Select-Object Name, PortBinding, VlanConfiguration, NumPorts

# Compare to the orphan DVPG VLAN (if dvportgroup-59 still exists in inventory by KEY)
Get-View -ViewType DistributedVirtualPortgroup |
    Where-Object { $_.MoRef.Value -in @( 'dvportgroup-59', 'dvportgroup-6025' ) } |
    Select-Object Name, @{N='VLAN';E={$_.Config.DefaultPortConfig.Vlan.VlanId}}
```

Gate: `pg-mgmt` and `pg-mgmt-ephemeral` both report VLAN ID matching the current orphan DVPG's VLAN. If VLANs differ, this is no longer a network-stack-transparent move — STOP and re-plan with IP changes. (Lab note: the entire management network is `192.168.1.0/24` flat per the original deployment, so VLAN identity is expected to be 0/untagged, but verify.)

5.4 Confirm SSH access to esxi04 (required for Phase 4 fallback)

```
Test-NetConnection -ComputerName esxi04.lab.local -Port 22
```

Gate: TCP 22 reachable. Root password verified out-of-band.

5.5 Snapshot the management VMs (rollback insurance)

Action: Right-click each of `sddc-manager`, `vcf-ops`, `fleet`, `vcenter` in vCenter UI → Snapshots → Take Snapshot. Name format: `pre-vDS-migration-2026-05-16`. Memory **excluded**. Quiesce **excluded** (avoid I/O pause on running workloads).

Why: Per the user's verified runbook for this lab, snapshots are the lowest-friction rollback when a network change traps a management VM in a bad state. Snapshots will be deleted after Phase 5 verification succeeds.

Gate: All four VMs show the pre-migration snapshot in Snapshot Manager.

6. Phase 1 — Migrate sddc-manager

Migrating `sddc-manager` first because its outage during the migration has the lowest blast radius — VCF Operations and Fleet continue to function during a brief SDDC Manager unavailability, and SDDC Manager has no UI dependencies on other management VMs.

6.1 Pre-migration check

```
# Confirm current binding
Get-VM sddc-manager | Get-NetworkAdapter |
  Select-Object Name, @{N='PortgroupKey';E={$_.ExtensionData.Backing.Port.PortgroupKey}},
                    @{N='SwitchUuid'; E={$_.ExtensionData.Backing.Port.SwitchUuid}}
# Confirm reachability
Test-Connection -ComputerName sddc-manager.lab.local -Count 2 -Quiet
```

Gate: Current binding is on the orphan SwitchUuid. Ping succeeds.

6.2 Migrate the NIC

```
$vm = Get-VM sddc-manager
$nic = Get-NetworkAdapter -VM $vm
Set-NetworkAdapter -NetworkAdapter $nic -PortGroup (Get-VDPortgroup -Name 'pg-mgmt') -Confirm:$false
```

Why `Set-NetworkAdapter -PortGroup`: This issues a `ReconfigVM_Task` to ESXi via vCenter; the actual port switch happens on the ESXi host hosting `sddc-manager`. Brief NIC blip (< 5 seconds in practice on this lab). [BROADCOM-DOC: KB 426300] documents this method for non-vCenter VMs.

6.3 Verify post-migration

```
# Re-read NIC binding
Get-VM sddc-manager | Get-NetworkAdapter |
  Select-Object @{N='PortgroupKey';E={$_.ExtensionData.Backing.Port.PortgroupKey}},
                @{N='SwitchUuid'; E={$_.ExtensionData.Backing.Port.SwitchUuid}}
# Connectivity
Test-Connection -ComputerName sddc-manager.lab.local -Count 4
# Service health
ssh vcf@sddc-manager.lab.local 'systemctl status lcm --no-pager | head -10'
```

Gate: `PortgroupKey=dvportgroup-45`, `SwitchUuid=50 12 35 19...`. Ping succeeds with ≤ 1 packet loss. `lcm` service shows `active (running)`.

Rollback if gate fails: Revert NIC to `dvportgroup-59` via `Set-NetworkAdapter -PortGroup (Get-VDPortgroup -Name '< orphan-dvpg>')` if the orphan DVPG is still in inventory by key — OR revert to the pre-migration snapshot from \$5.5.

7. Phase 2 — Migrate vcf-ops

Same procedure as Phase 1 against `vcf-ops` . Migrating second because VCF Operations UI is the primary admin path for Fleet — moving Fleet (Phase 3) is easier with Ops fully reachable.

7.1 Migrate

```
$vm = Get-VM vcf-ops
$nic = Get-NetworkAdapter -VM $vm
Set-NetworkAdapter -NetworkAdapter $nic -PortGroup (Get-VDPortgroup -Name 'pg-mgmt') -Confirm:$false
```

7.2 Verify

```
Get-VM vcf-ops | Get-NetworkAdapter |
  Select-Object @{N='PortgroupKey';E={$_.ExtensionData.Backing.Port.PortgroupKey}}
Test-Connection -ComputerName vcf-ops.lab.local -Count 4
# UI reachable
Invoke-WebRequest -Uri 'https://vcf-ops.lab.local/' -UseBasicParsing -SkipCertificateCheck -TimeoutSec 15
|
  Select-Object StatusCode
```

Gate: `PortgroupKey=dvportgroup-45` . UI returns HTTP 200 (or 302 to login). Wait up to 90 seconds for `vmware-vcops-web` to reconnect to its peer services if it stutters.

8. Phase 3 — Migrate fleet

Same procedure against `fleet` .

8.1 Migrate

```
$vm = Get-VM fleet
$nic = Get-NetworkAdapter -VM $vm
Set-NetworkAdapter -NetworkAdapter $nic -PortGroup (Get-VDPortgroup -Name 'pg-mgmt') -Confirm:$false
```

8.2 Verify

```
Get-VM fleet | Get-NetworkAdapter |
  Select-Object @{N='PortgroupKey';E={$_.ExtensionData.Backing.Port.PortgroupKey}}
Test-Connection -ComputerName lcm.lab.local -Count 4
# Fleet appliance:8080 returns its banner (HTTP 401 is expected – see [[reference-vcf9-fleet-access-pattern]])
Invoke-WebRequest -Uri 'https://lcm.lab.local:8080/' -UseBasicParsing -SkipCertificateCheck -TimeoutSec 15
-ErrorAction SilentlyContinue
```

Gate: `PortgroupKey=dvportgroup-45` . Ping succeeds. Fleet 8080 responds (any HTTP code other than connection-refused).

9. Phase 4 — Migrate vCenter (Special Case)

`vcenter` is the management plane being moved; it cannot reliably issue its own `ReconfigVM_Task` to itself without losing the very connection that completes the task. The documented procedure for this case is [\[BROADCOM-DOC: KB 42](#)

6300, KB 418461] : add a second NIC on the ephemeral DVPG, verify connectivity through it, then remove the original NIC.

9.1 Method A (recommended) — Dual-NIC via vCenter UI [BROADCOM-DOC: KB 426300]

Step 9.1.1 — Add second NIC bound to `pg-mgmt-ephemeral`

In vCenter UI: `vcenter` VM → Edit Settings → ADD NEW DEVICE → Network Adapter → select `pg-mgmt-ephemeral` → Adapter Type: VMXNET 3 → Status: Connect at Power On + Connected.

Or via PowerCLI:

```
$vm = Get-VM vcenter
$dpg = Get-VDPortgroup -Name 'pg-mgmt-ephemeral'
New-NetworkAdapter -VM $vm -Portgroup $dpg -Type Vmxnet3 -StartConnected:$true -Confirm:$false
```

Step 9.1.2 — Configure the new NIC inside vCenter (Photon OS)

SSH to vCenter as `root` and configure `eth1` (or whichever interface the new NIC enumerates as):

```
# Identify the new interface (will be the one without an IP)
ip -br link show

# Bring it up (Photon uses systemd-networkd)
cat > /etc/systemd/network/20-eth1.network <<'EOF'
[Match]
Name=eth1

[Network]
DHCP=no
LinkLocalAddressing=no
EOF

networkctl reload
ip link set eth1 up
```

Do **not** assign an IP yet — at this point the interface only needs to be UP so the underlying ESXi-host port assignment completes.

Step 9.1.3 — Verify the second NIC connects

From the ESXi host hosting vCenter (`esxi04`):

```
esxcli network vm list # find world ID for 'vcenter'
esxcli network vm port list -w <worldId>
```

Gate: Two ports listed — one on the orphan switch, one on `pg-mgmt-ephemeral`. Both `Active`.

Step 9.1.4 — Move vCenter's IP to the new NIC

In Photon, swap the `192.168.1.69` IP from `eth0` to `eth1`:

```
# Capture current eth0 config (review before applying)
cat /etc/systemd/network/10-eth0.network

# Update eth1 to take over the management IP
cat > /etc/systemd/network/20-eth1.network <<'EOF'
[Match]
```

```

Name=eth1

[Network]
DHCP=no
Address=192.168.1.69/24
Gateway=192.168.1.1
DNS=192.168.1.230
DNS=8.8.8.8
EOF

# Disable eth0 by renaming its config so it does not bring an IP up
mv /etc/systemd/network/10-eth0.network /etc/systemd/network/10-eth0.network.disabled

networkctl reload
# Bring eth0 down and eth1 up – order matters
ip addr flush dev eth0
ip link set eth0 down
ip link set eth1 up

```

Why this order: The Address line on `eth1` claims `192.168.1.69` while `eth0` still holds it; `systemd-networkd` does not auto-renumber. The flush + link-down on `eth0` is what forces the address transition. Expect a 1-3 second connectivity gap; vCenter UI session may drop and need re-login. ARP cache on upstream switches may take up to 30 seconds to update — pings from the workstation will recover within that window.

Step 9.1.5 — Verify connectivity via new NIC

```

Test-Connection -ComputerName vcenter.lab.local -Count 6
# Re-login to vCenter UI
Connect-VIServer vcenter.lab.local
Get-VM vcenter | Get-NetworkAdapter |
    Select-Object Name, NetworkName,
        @{N='PortgroupKey';E={$_.ExtensionData.Backing.Port.PortgroupKey}},
        @{N='SwitchUuid'; E={$_.ExtensionData.Backing.Port.SwitchUuid}}

```

Gate: Two NICs listed — `eth0` (orphan, IP not assigned) and `eth1` (`pg-mgmt-ephemeral`, IP `192.168.1.69`). vCenter UI reachable. PowerCLI `Connect-VIServer` succeeds.

Step 9.1.6 — Remove the orphan NIC

```

$vm = Get-VM vcenter
Get-NetworkAdapter -VM $vm | Where-Object {
    $_.ExtensionData.Backing.Port.SwitchUuid -eq '50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8'
} | Remove-NetworkAdapter -Confirm:$false

```

Gate: `Get-VM vcenter | Get-NetworkAdapter` returns exactly one NIC, on `pg-mgmt-ephemeral`.

9.2 Method B (fallback) — In-place swap via direct-to-ESXi connection

Use only if Method A fails or the dual-NIC IP transition is judged too disruptive.

```

# Disconnect any session with vCenter so we don't hairpin through it
Disconnect-VIServer -Server vcenter.lab.local -Confirm:$false -Force

# Connect directly to the host running vCenter
Connect-VIServer -Server esxi04.lab.local -User root -Password '<root pwd>'

# Swap the NIC – note that target portgroup must be referenced by VDPortgroup name,
# but on a direct ESXi connection we have to use the standard PortGroup binding
$vm = Get-VM vcenter
$nic = Get-NetworkAdapter -VM $vm

```

```
Set-NetworkAdapter -NetworkAdapter $nic -PortGroup (Get-VDPortgroup -Name 'pg-mgmt-ephemeral') -
Confirm:$false
```

Risk: [BROADCOM-DOC: KB 418461] documents that this approach can leave vCenter inaccessible if the new DVPG uses static binding (vCenter cannot assign its own port). This plan migrates **to an ephemeral DVPG** specifically to avoid that failure mode — ephemeral binding lets the ESXi host assign the port directly without vCenter mediation, which is the exact scenario KB 418461 prescribes.

Recovery if Method B leaves vCenter unreachable: Revert via the pre-migration snapshot (\$5.5). Or, from ESXi host CLI:

```
# Find the vCenter VM
vim-cmd vmsvc/getallvms | grep -i vcenter

# Use the vSphere MOB on the ESXi host (port 443) to reissue ReconfigVM_Task with
# a fallback portgroup. Documented in [BROADCOM-DOC: KB 318719].
```

10. Phase 5 — Post-Migration Verification

Goal: prove all four migrations took and that NSX cleanup is now possible.

10.1 All four VMs on live vDS

```
Get-VM sddc-manager, vcf-ops, fleet, vcenter |
  ForEach-Object {
    $v = $_
    Get-NetworkAdapter -VM $v | ForEach-Object {
      [PSCustomObject]@{
        VM           = $v.Name
        PortgroupKey = $_.ExtensionData.Backing.Port.PortgroupKey
        SwitchUuid   = $_.ExtensionData.Backing.Port.SwitchUuid
      }
    }
  } | Format-Table -AutoSize
```

Gate: All four rows show SwitchUuid = 50 12 35 19 fa f5 be 0a-71 8d ab e0 b5 e6 b9 cf (live). sddc-manager, vcf-ops, fleet show PortgroupKey=dvportgroup-45 (pg-mgmt). vcenter shows PortgroupKey=dvportgroup-50 (pg-mgmt-ephemeral).

10.2 Re-run NSX ownership-clear (the gate that failed earlier)

```
$cmId    = '259b8edb-3b54-496f-bfb4-c85c145cd9bb'
$oldVds  = '50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8'
$url     = "https://192.168.1.71/api/v1/managed-objects/switching/nsx-ownership/$cmId" + "?action=clear"
$body    = @{ mo_id = $oldVds } | ConvertTo-Json
Invoke-WebRequest -Uri $url -Method Put -Headers $h -Body $body -UseBasicParsing
```

Gate: HTTP 200, no error_code 8012. If still 8012, the orphan vDS still has live references somewhere — re-run §10.1 and identify which VM was missed.

10.3 Run the lab health check

```
cd "/i/VCF Health check"
```

```
bash vcf-health-check.sh
```

Gate: Report grade B or better. Zero new FAILs introduced. Existing yellow alarms (vSAN latency, Cluster Configuration Consistency) are pre-existing and acknowledged.

10.4 Confirm HB-1 / HB-2 satisfied

```
Get-VM vcenter | Get-NetworkAdapter | ForEach-Object {  
    $key = $_.ExtensionData.Backing.Port.PortgroupKey  
    $dpg = Get-VDPortgroup | Where-Object { $_.Key -eq $key }  
    "HB-1 (DVS portgroup)      : $($dpg.Name)          (must NOT be standard vSwitch)"  
    "HB-2 (Ephemeral)         : $($dpg.PortBinding)    (must be Ephemeral)"  
}
```

Gate: Output reports `pg-mgmt-ephemeral` and `Ephemeral`. HB-1 and HB-2 both pass.

10.5 Remove pre-migration snapshots

After §10.1 - §10.4 pass and **at least 60 minutes** of clean operation:

```
Get-VM sddc-manager, vcf-ops, fleet, vcenter |  
Get-Snapshot -Name 'pre-vDS-migration-2026-05-16' |  
Remove-Snapshot -Confirm:$false
```

11. Phase 6 — Orphan vDS Removal (Optional)

Run this phase only **after** Phase 5 fully passes and **only with explicit user authorization** as a separate change. Out of scope for the same-day execution by default.

11.1 Confirm zero VMs on orphan vDS

```
Get-VM | ForEach-Object {  
    $v = $_; Get-NetworkAdapter -VM $v | ForEach-Object {  
        if ($_.ExtensionData.Backing.Port.SwitchUuid -eq '50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8') {  
            "$($v.Name) still bound to orphan!"  
        }  
    }  
}
```

Gate: Empty output.

11.2 Confirm zero hosts still uplinked to orphan vDS

From `esxi01.lab.local`:

```
esxcli network vswitch dvs vmware list
```

Gate: Output shows only `vcenter-cl01-vds01`. If the orphan UUID still appears, the host uplinks need detachment first — that is the NSX-cleanup work that has been blocked until this plan completes, and is the next document.

11.3 Remove the orphan vDS

If the orphan switch appears in `Get-VDSwitch`:

```
Get-VDSwitch | Where-Object { $_.ExtensionData.Uuid -eq '50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8' } | Remove-VDSwitch -Confirm:$false
```

If it does not appear (ghost reference only — host-level `vmware dvs` records but no vCenter view), Broadcom does not document a supported removal path via PowerCLI; the cleanup runs as part of the NSX TNC detach/reattach which is itself unblocked after this plan completes.

12. Risk Register

ID	Risk	Likelihood	Impact	Mitigation
R-1	VM loses connectivity during NIC swap	Low (same VLAN)	Medium (mgmt VM offline 5-30s)	Hot-swap on identical VLAN should keep IP and ARP stable; snapshots provide rollback
R-2	vCenter session drops during its own NIC change	Medium (Method B) / Low (Method A)	High (cannot manage cluster)	Default to Method A (dual-NIC, KB 426300); if vCenter unreachable, revert snapshot
R-3	Target ephemeral DVPG (<code>pg-mgmt-ephemeral</code>) has zero ports	Low — <code>NumPorts=0</code> is normal for ephemeral DVPGs by design	None	Ephemeral allocates on demand; no action
R-4	Migration triggers vSphere DRS-driven vMotion of a moved VM	Low (4 hosts, all healthy)	Low — vMotion is transparent	DRS = FullyAutomated may move VMs during/after; this is benign because all hosts are now on the live vDS at the host level
R-5	Orphan DVPG <code>dvportgroup-59</code> is deleted from inventory mid-migration	Low	High — would lose ability to revert VM 1 if VM 2 fails	Do all four migrations within a single window; do not pause overnight
R-6	NSX detects the topology change and re-emits new alarms	Medium	Low (informational)	Expected — these alarms will replace the orphan-vDS alarms and self-clear once Phase 6 (or downstream NSX cleanup) completes

13. Rollback Procedures

13.1 Single-VM rollback (Phases 1-3)

If a non-vCenter migration causes the VM to become unreachable:

```
# Option A: NIC revert (only if orphan DVPG still in inventory by key)
$vm = Get-VM <vm-name>
$nic = Get-NetworkAdapter -VM $vm
Set-NetworkAdapter -NetworkAdapter $nic -PortGroup (Get-VDPortgroup | Where-Object Key -eq 'dvportgroup-59') -Confirm:$false
```

```
# Option B: Snapshot revert
Get-VM <vm-name> | Get-Snapshot -Name 'pre-vDS-migration-2026-05-16' | Set-VM -Snapshot $_ -Confirm:$false
```

13.2 vCenter rollback (Phase 4 Method A)

If vCenter is unreachable after Step 9.1.4 (IP swap to new NIC):

1. Direct SSH to vCenter via console (vSphere Web Console from another VM, or DCUI on `esxi04.lab.local` → console → power → `vcenter`)
2. Revert the network config:

```
bash mv /etc/systemd/network/10-eth0.network.disabled /etc/systemd/network/10-eth0.network rm /etc/systemd/network/20-eth1.network networkctl reload ip addr flush dev eth1 ; ip link set eth1 down ip link set eth0 up
```

3. If still unreachable, revert pre-migration snapshot.

13.3 vCenter rollback (Phase 4 Method B)

If in-place NIC swap leaves vCenter unreachable:

1. Revert the pre-migration snapshot directly from `esxi04` :

```
bash vim-cmd vmsvc/getallvms | grep vcenter vim-cmd vmsvc/snapshot.get <vmid> # confirm snapshot present vim-cmd vmsvc/snapshot.revert <vmid> <snapshotId> 0
```

2. After revert completes, verify connectivity, then return to Method A.

14. KB References

KB	Title	Applies to
[BROADCOM-DOC: KB 324492]	Static (non-ephemeral) or ephemeral port binding on a vSphere Distributed Switch	HB-2 background; binding semantics
[BROADCOM-DOC: KB 426300]	Moving vCenter Server to an ephemeral port group on a vSphere Distributed Switch	Phase 4 Method A — canonical procedure
[BROADCOM-DOC: KB 418461]	vCenter Server becomes inaccessible after switching the network adapter on the VC VM from a portgroup on a standard switch to one on a virtual distributed switch	Phase 4 failure mode and dual-NIC pattern
[BROADCOM-DOC: KB 318719]	vCenter network connectivity lost — Recover vCenter network when connected to a Distributed Switch	Rollback procedures
[BROADCOM-DOC: KB 428279]	Inability to Enable EVC on Cluster Containing vCenter Server Due to "Non-Ephemeral Distributed Virtual Port Groups" Error	HB-2 supporting evidence
[BROADCOM-DOC: KB 422607]	Unable to connect a newly deployed vCenter appliance VM to an existing distributed port group	Related — fresh VC deploy edge case
[BROADCOM-DOC: KB 413513]	Unable to add distributed portgroup for a VM's network adaptor in ESXi host client where ESXi host is in disconnected state in vCenter	Edge case if host disconnects mid-migration
[BROADCOM-DOC: legacyId=81611]	VMware Cloud Foundation — Design Decision — Use ephemeral port binding for the management port group	Authoritative basis for HB-2

Companion documents in this repo:

- `VCFA-Deploy-Airtight-Plan-2026-05-12.md` — HB-1 / HB-2 / HB-3 / HB-4 source
- `NSX-Trust-CM-TNP-Repair-Plan-2026-05-15.md` — the cleanup that this plan unblocks
- `reference_nsx9_tnp_orphan_vds_repair.md` (memory) — the per-host TN error path
- `reference_vcf9_fleet_access_pattern.md` (memory) — Fleet probe expectations

Appendix A — Commands

```
# === Pre-flight (Phase 0) ===
Connect-VIServer vcenter.lab.local
Get-VDSwitch | Select Name, @{N='UUID';E={$_.ExtensionData.Uuid}}, Version
Get-VM | ForEach-Object {
    $v=$_; Get-NetworkAdapter -VM $v | ForEach-Object {
        [PSCustomObject]@{ VM=$v.Name; PortgroupKey=$_.ExtensionData.Backing.Port.PortgroupKey;
        SwitchUuid=$_.ExtensionData.Backing.Port.SwitchUuid }
    }
} | Format-Table -AutoSize
Get-VDPortgroup -Name pg-mgmt, pg-mgmt-ephemeral | Select Name, PortBinding, VlanConfiguration, NumPorts

# Snapshots
Get-VM sddc-manager, vcf-ops, fleet, vcenter | ForEach-Object {
    $_ | New-Snapshot -Name 'pre-vDS-migration-2026-05-16' -Memory:$false -Quiesce:$false -Confirm:$false
}

# === Migration (Phases 1-3) ===
foreach ($n in 'sddc-manager','vcf-ops','fleet') {
    $vm = Get-VM $n
    $nic = Get-NetworkAdapter -VM $vm
    Set-NetworkAdapter -NetworkAdapter $nic -PortGroup (Get-VDPortgroup -Name 'pg-mgmt') -Confirm:$false
    Start-Sleep 5
    Test-Connection -ComputerName $n -Count 4
}

# === vCenter Method A (Phase 4) ===
# Step 9.1.1
$vm = Get-VM vcenter
$dpg = Get-VDPortgroup -Name 'pg-mgmt-ephemeral'
New-NetworkAdapter -VM $vm -Portgroup $dpg -Type Vmxnet3 -StartConnected:$true -Confirm:$false

# Inside vCenter Photon shell – Step 9.1.2 through 9.1.4 (see body)

# Step 9.1.6 – remove orphan NIC
Get-NetworkAdapter -VM (Get-VM vcenter) | Where-Object {
    $_.ExtensionData.Backing.Port.SwitchUuid -eq '50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8'
} | Remove-NetworkAdapter -Confirm:$false

# === Post-migration verification (Phase 5) ===
Get-VM sddc-manager, vcf-ops, fleet, vcenter | ForEach-Object {
    $v=$_; Get-NetworkAdapter -VM $v | ForEach-Object {
        [PSCustomObject]@{ VM=$v.Name; PortgroupKey=$_.ExtensionData.Backing.Port.PortgroupKey;
        SwitchUuid=$_.ExtensionData.Backing.Port.SwitchUuid }
    }
} | Format-Table -AutoSize

# Re-run NSX ownership-clear
$basic = [Convert]::ToBase64String([Text.Encoding]::UTF8.GetBytes('admin:<LAB_PW>'))
$h = @{ Authorization="Basic $basic"; Accept='application/json'; 'Content-Type'='application/json' }
$cmId = '259b8edb-3b54-496f-bfb4-c85c145cd9bb'
$body = @{ mo_id='50 37 13 3f 96 e4 3e 93-1d f0 3b 01 d4 76 6a f8' } | ConvertTo-Json
Invoke-WebRequest -Uri "https://192.168.1.71/api/v1/managed-objects/switching/nsx-ownership/$cmId?action=clear" -Method Put -Headers $h -Body $body -UseBasicParsing
```

Document Information

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End of plan.